

Step-by-Step Guide: Deploying Streamlit App on Streamlit Cloud (using GitHub)

1. Prepare Your Project Folder

Your project folder should contain the following essential files:

```
my_streamlit_app/
|
├─ app.py                # Your main Streamlit application file
├─ requirements.txt       # List of all Python packages with versions
├─ best_model.pkl         # Your trained ML model or pipeline
└─ any_other_files/      # (Optional) assets, data files, etc.
```

Example `requirements.txt`

```
streamlit==1.39.0
pandas==2.2.2
numpy==1.26.4
scikit-learn==1.5.1
joblib==1.4.2
```

Tips

- Always pin specific versions to avoid compatibility issues.
- Make sure your `.pkl` model and any other necessary files are in the same folder as `app.py` or properly referenced in code.

2. Verify Locally Before Uploading

Run your Streamlit app locally to confirm it works:

```
streamlit run app.py
```

- If it opens correctly in your browser, you're good to go.
-

3. Upload Project to GitHub

1. Create a **new GitHub repository** (public).
2. Add all files (`app.py`, `requirements.txt`, model file, etc.).
3. Commit and push your code

4. Deploy on Streamlit Cloud

1. Go to <https://streamlit.io/cloud>
 2. Click “**Sign in with GitHub**” and authorize access.
 3. Click “**Create app**” → “**Deploy from GitHub**”.
 4. Select your repository and branch (`main`).
 5. Set the **Main file path** to your Streamlit script (e.g., `app.py`).
 6. Click “**Deploy App**” 
-

5. Streamlit Cloud Builds Automatically

- Streamlit Cloud automatically installs all dependencies from **requirements.txt**.
- Once deployment completes, your app will be live with a public URL like:

`https://<your-app-name>.streamlit.app`

Optional (If Using Pickled Model)

- Ensure your **.pkl** file was saved using compatible versions of libraries (e.g., same **scikit-learn** version).

When loading in Streamlit:

```
import pickle
model = pickle.load(open('best_model.pkl', 'rb'))
```
